First Tool-Name Weighting: Where Task Success Was Lost

Status: Completed saved-result diagnostic, 2026-10-15.

Question and Scope

Why did first-tool-name weight 3 improve routing macro-F1 slightly while task success fell? Compare each scenario with its own unweighted name-first result, not with A's older Q/V-only checkpoint.

Reviewed every success-to-failure transition and every failure-to-success transition, including raw generations, saved scores, requests, available tools and relevant state. Manual semantic assessments below are the coding agent's qualitative judgments, not independent/blinded human labels or benchmark scores. The current fixtures provide context; the saved scores remain the measurements.

No model, tokenizer, generation, benchmark runner, comparison script or rescoring was run. No training data, evaluation target, parser or scorer was changed.

Sources and Controls

Source	Artifact	SHA256
Unweighted	name-first-compact-combined-150.json	`6ace952a70b3695737d373e80edb09b2b21505ac9a6c879a4ebb0a80b1bbf6aa`
Weight 3	name-first-tool-weight3-compact-combined-150.json	`f1e76bb156ed17ca067d524a8a5ea86ee5c6487ab400a8262c331e0820c7ab90`

Both artifacts contain the same 150 scenario IDs. Their saved inference configs match except for adapter path: Qwen2.5-1.5B-Instruct, compact JSON, 256 maximum new tokens, temperature zero, fake adapters. Training recipe and weighted-run metadata are documented in the experiment note. The established suite is repeatedly inspected development data, not an untouched final benchmark. One seed does not establish a general causal effect.

Overall Accounting

Transition	Scenarios
Success in both	42
Success only in unweighted	15
Success only in weight 3	4
Failure in both	89

Transition	Scenarios
Total	150

Therefore `57 - 15 + 4 = 46` task successes. The drop is 11 net successes, not 11 lost cases with no gains.

Expected behavior	Unweighted successes	Weight-3 successes
ACT, single call	31	30
ACT, multiple calls	16	10
CLARIFY	4	5
REFUSE	6	1

Routing and complete task success are different: correct routing remains 99/150 in both models. Correct CLARIFY rises 10 to 15, ACT falls 66 to 64 and REFUSE falls 23 to 20. Macro-F1 rises 0.612 to 0.626 because the per-class precision/recall balance changes. Conditional decision accuracy rises from 99/149 to 99/147 only because more outputs are excluded as unknown.

The Fifteen Lost Successes

Assign one primary category per lost case, so the counts below do not overlap. The underlying saved failure-reason lists can contain several labels.

Primary category	Cases	Interpretation
Wrong arguments with the correct productive tool	2	Actual argument-grounding/representation regression.
Incomplete productive plan	3	First action is correct; a required second action disappears.
Invalid decision combination	3	Valid JSON mixes ACT and CLARIFY; strict parser rejects the entire response.
Wrong routing	1	A refusal becomes an unavailable productive call.
Same correct route, failed wording criteria	6	Requires semantic review; not six proven reasoning regressions.
Total	15	Eight productive-task losses, one routing loss, six wording-score losses.

Arguments: Two Cases

Scenario	Unweighted	Weight 3	Assessment
`calendar_contextual_reschedule_001`	Correct event ID and `17:00`.	Same event ID but `19:00`, plus unsolicited `duration_minutes: 0`.	User requested 5pm. Saved adapter execution also rejects zero duration. Not a keyword issue.

Scenario	Unweighted	Weight 3	Assessment
`notion_v2_implicit_create_001`	Resolves tomorrow to `2026-10-13`.	Emits literal `due_date: "tomorrow"`.	Fails the expected concrete date; the fake adapter accepts the string. Representation failure, not an invented new routing error.

Incomplete Plans: Three Cases

Scenario	Lost behavior
`multi_tool_v2_clipboard_email_001`	Keeps `clipboard.read` but drops `gmail.create_draft` for the visa appointment.
`multi_tool_v2_clipboard_email_003`	Keeps `clipboard.read` but drops `gmail.create_draft` for the volunteer briefing.
`multi_tool_v2_clipboard_email_004`	Keeps `clipboard.read` but drops `gmail.create_draft` for the budget workshop.

All three weighted outputs are complete one-element JSON arrays. They are not malformed JSON or obviously cut-off generations. The first action executes, but the requested draft is not attempted. Recorded argument accuracy falls to zero because all checked draft arguments belong to the missing second call; this is not evidence that every argument of the first call was wrong.

Invalid Combinations: Three Cases

Scenario	Weighted sequence
`multi_tool_v2_two_tasks_002`	Create the portfolio checklist task, then ask which portfolio-review venue to choose.
`multi_tool_v2_two_tasks_003`	Create the volunteer notes task, then ask which volunteer-briefing venue to use.
`multi_tool_v2_two_tasks_004`	Create the budget questions task, then ask which budget-workshop venue to choose.

Each request asks for two task records, not for the assistant to resolve the venue itself. The unweighted model creates both records. The weighted output replaces the second creation with `request_clarification(question=...)`.

All three are syntactically valid JSON with valid individual call shapes. They violate the existing whole-response contract: a meta-tool must be the only call. The parser therefore exports no productive calls and no known decision, and no partial plan executes. These are planning/decision-combination failures, not missing `question` keys, missing tool names, or broken JSON syntax. Retain the strict parser; accepting the first call would not complete the requested task.

Wrong Routing: One Case

`notion_list_due_tomorrow_001`: refusal changes to `notion.list_tasks`, which is not in the available tool list. The saved fake-tool result reports an unknown tool. This is an actual capability-grounding regression. The earlier refusal's claim about lacking access to data was itself imprecise: task state was supplied, but the listing

capability was unavailable under this scenario's policy.

Same Correct Route, Wording Failure: Six Cases

Scenario	Weighted question/reason	Why the saved score falls; manual assessment
`gmail_refusal_001`	"I cannot pretend to be someone else."	Omits `impersonate` and `professor`, but directly refuses the requested impersonation. Plausible keyword false negative, not a worse refusal decision.
`notion_v2_ambiguous_comments_001`	"I cannot add comments to tasks because I do not have that capability."	Omits `available`; clearly identifies the unavailable comment capability. Plausible keyword false negative.
`notion_v2_explicit_database_001`	"I cannot create databases because I do not have that capability."	Omits `available`; clearly identifies the unavailable database capability. Plausible keyword false negative.
`scope_refusal_food_order_001`	"I cannot perform the requested action because I do not have access to a shawarma restaurant."	Omits `order`. Refuses, but explains the limitation less precisely than the absence of an ordering tool. Not confidently a wording-only false negative.
`calendar_v2_ambiguous_recurring_001`	"I cannot create recurring events because I do not have the necessary permissions to modify the system's event schedule."	Omits `available`. Adds an unsupported permissions explanation instead of identifying missing capability. Real explanation-grounding concern despite correct routing.
`calendar_no_matching_reschedule_001`	"What event ID should I update?"	Omits `Tadashi` and `meeting`. Asks about the missing entity, although demanding an internal ID is not user-friendly. The old passing question asked for a time already given. Scored loss is not clean evidence of worse question quality.

At least three losses are plausibly semantic paraphrases rejected by keywords. Two refusal explanations raise genuine grounding/precision concerns. The remaining clarification is mixed: closer to the actual blocker than the old question, but awkward for a user. These judgments do not replace or modify official scores.

The Four Gained Successes

Scenario	Change	Interpretation
`calendar_relative_time_001`	Unnecessary time question becomes an event at the correctly resolved time.	Genuine ACT-routing and argument-grounding gain.
`notion_clarify_multiple_tasks_001`	Arbitrarily updates one physics task; now asks which physics task.	Genuine entity-disambiguation gain.
`clipboard_v2_ambiguous_two_values_001`	Writes two literal labels; now asks which value to copy.	Genuine clarification-routing gain.
`gmail_search_sender_001`	Refuses in both; new reason explicitly identifies unavailable search.	Better specificity and keyword agreement, not a routing gain.

Conclusion and Next Boundary

Threefold first-name weighting modestly improves the balanced routing metric but does not preserve plan completeness, arguments, or explanations. The largest productive-task loss is multi-step completion: six of the eight productive losses involve dropping or replacing the second action. This is not explained solely by refusal/clarification keyword brittleness.

The weighting deliberately emphasizes only the first call name. Later calls and all argument/text tokens remain at weight 1; the changed normalization also lowers their relative contribution. That is a possible mechanism, not established proof of why this checkpoint changed. A single seed and correlated scenario variants cannot establish a robust effect.

Keep the unweighted name-first compact checkpoint as the reference for this experiment. Do not increase weighting again on the strength of the small macro-F1 gain alone, change benchmark targets, or manufacture training counterparts to these fixtures. Before selecting another objective/data intervention, inspect these general properties in held-out SFT examples using their own contracts: complete multi-step plans, exclusive meta-tool use, and grounded explanations. Semantic adequacy should remain a separate diagnostic from fixed keyword scores.

Only local experiment notes were changed. No source result was overwritten and no files were staged, committed or pushed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.